

persist for days at a time, or longer. This can be prevented by suitable maintenance measures. On the other hand with the grounded-neutral system the tendency is for such accidental contacts to develop into short circuits and to operate the circuit breakers. From the discussion by Messrs. Crawford and Cunningham it is evident that the local conditions immediately surrounding an accidental ground contact cause great variations in contact resistance and resulting ground current.

Mr. Crawford has raised the question of joint use at higher voltages than has been customary in the past. Many of you are doubtless aware that the subject of conditions of joint use is under active study by the Joint General Committee of the N. E. L. A. and the Bell Telephone System. We do, of course, recognize that there are some special cases where joint use of poles at higher voltages is the best engineering solution to meet the conditions of a particular problem and in such cases joint use is being approved.

Mr. Crawford also stated that his company had lately acquired a combination power and telephone distribution system with the power lines operating at 6600 volts. Our experience has not been encouraging as to the conditions of noise and hazard that obtain in such circumstances.

There are, of course, differences of opinion in respect to the question of protection from the hazards of power distribution. I wish to suggest, first, that higher voltages are ordinarily employed in distribution for longer distances and larger loads and, second, that this inevitably means either extraordinary measures to prevent hazard and impairment of service or else lowered standards. It is our view that advance planning will enable us to avoid to a great extent the necessity of joint use at the higher voltages. Meanwhile, any specific situation that arises we are more than ready to consider.

Mr. Corbett has described the working out of cooperative consideration of specific cases in California. As he states, the best results are obtained when the reactions of proposed measures are thoroughly considered by both parties. Suggestions arising from the study of these problems by the power engineers are cordially welcomed. I think it fair to say that the working together in California is not merely a matter of compliance with regulations but a realization that it is the rational way for the power and communication utilities to solve these problems. Mr. Mayer's discussion brings out the fact that there are conflicts to be adjusted between the requirements of the communication and power distributions and that the degree of detriment from multiple grounding depends upon several factors in the layout of the systems. Mr. Mayer's point about the use of multiple grounds on the primary neutral for stabilization has also been brought out by Mr. Hood. While recognizing that accurate balance of loads might prevent detrimental induction in communication circuits so long as all phases are present, we must not overlook the facts that setting up and continuously maintaining such close balance is not a simple matter and that exposures often occur where only one- or two-phase wires are involved.

OPPORTUNITIES AND PROBLEMS IN THE ELECTRIC DISTRIBUTION SYSTEM¹

(BLAKE)

SEATTLE, WASHINGTON, SEPTEMBER 18, 1925

C. E. Carey: I wish to discuss one or two details in Mr. Blake's paper. He mentions primarily the single-pole protective unit. He goes into detail and attempts to establish the justification of the multiplicity of parts. However, the question of single-phase or single-pole units cannot be separated from the whole network. Personally, I believe that the multiple unit is the real solution. We have seen that there are certain state laws requiring that we open the entire circuit, rather than

allow an unbalance on the networks and produce inductive interference. It is much better, I believe, to clear the entire trouble as quickly as possible rather than take out one phase at a time. The three-pole unit will not occupy as much space in the manhole as three single-pole units. The automatic a-c. network protector is a device which does everything you can expect it to do. It is designed to go into the standard 35-in. manhole cover. The covers of the units are of aluminum, are light and can be handled by one man. Furthermore, if inspection is to be made of these devices the time consumed will be less with three-pole units than with three single-pole units.

In regard to the translator, I believe, that anything we put in to complicate the network is defeating the purpose of the a-c. system. The whole thing resolves itself back into continuity of service, simplicity and low cost, and the only way we can arrive at that service with a reasonable cost is to trim it down to just the bare necessities.

Practically everyone will agree that if we had it to do over again, we would never put in a d-c. network. We have it, but what are we going to do with it? If we attempt to change to an a-c. network we are confronted with the cost of changing over the customer's equipment, which will be so high in some cases that the interest and other fixed charges on that additional cost will more than carry the high losses in the d-c. network. However, one solution is being used to bring the economics of the d-c. very close to those of the a-c. network, the use of the automatic substation. The automatic substation gives primarily the same type of distribution as the a-c. system; that is, high voltages to the converting stations and low-voltage for distribution. The automatic substation is reliable and economical. A real analysis of the problem will often show that it is better to hold to the d-c. network, remove the concentrated manually operated stations, and distribute that conversion power over a network to a large number of small conversion stations, approaching extremely close to the transformers on the a-c. network in losses and other factors which come into the distribution problem.

Henry Richter: In considering a-c. secondary networks, it is well to recognize that the network systems mentioned by the United Electric Light and Power Co. as used in New York City—and also those in New Orleans—use triple-pole automatic network units exclusively, and that similar systems and network units will shortly be in operation in Dallas, Memphis, Knoxville, and Atlanta. Minneapolis will install these units as the load grows, preparatory to forming a network. These comprise all the companies that have completed or started to construct a network like that of Fig. 2, using automatic network units.

In practically none of the installations in these cities, where space might permit single-pole units to be installed, has it been necessary to add to the dimensions of new manholes in order to accommodate the triple-pole network unit; or to construct new manholes, or enlarge old ones, due to any extra space required because single-pole units were not available; or even to take up space in an existing vault that could be ill spared. These networks are considered justifiable only in heavily loaded areas where the loads are too great to be supplied in the single-phase manner. Three-phase transformer banks are therefore the rule, ranging in size from three 25-kv-a. single-phase transformers up to three 100-kv-a. transformers. The manholes and building vaults to accommodate such banks, to allow for growth of load, to ensure no excessive temperatures, and to permit proper racking and maintenance of cables, must be of such size that there is no lack of wall space for a triple-pole network unit. It is customary to locate this unit on the wall opposite that along which the three transformers are placed, and close to a corner of the manhole. In existing manholes having three-phase banks formerly part of a radial primary and secondary system, the triple-pole network units can usually

¹ Abridged in A. I. E. E. JOURNAL, December, 1925, p. 1355. Complete copies in pamphlet form only.

be put into the space formerly occupied by the large secondary junction boxes. In most cases, this space is not otherwise useful, for the network systems employing these automatic network units are so simple that nothing else is necessary in the manhole besides the transformers with or without small reactors, the cables and the network unit. Further, it is an erroneous idea that single-pole network units, particularly in a submersible housing, can be made so tiny that they can be installed in any odd corners of a manhole. The desirability of giving them proper maintenance makes such procedure far from advisable even if they were so small.

If it is desired to have the leads to the secondary mains leave the triple-pole network unit horizontally, to either side, it is extremely simple to provide a small terminal box where the outgoing leads emerge from the top of the submersible housing. This small box can have wiping nipples mounted horizontally at either side. The extra bend in the cable is thus eliminated. However, in none of the cities enumerated has this been necessary. Seven and a half feet is an average height for these transformer manholes as determined by good subway construction practise and even with a triple-pole network unit for a bank of three 100-kv-a. transformers there is no cramping of cables entering and leaving the housing.

When the triple-pole network unit was being designed, it was recognized that the submersible housing must pass through a certain minimum-size manhole opening. A country-wide survey was made and it was learned that the great majority of companies have adopted 35-in. diameter round as the smallest opening for transformer manholes in existing or contemplated construction. Apparently this minimum was governed by the dimensions of 100-kv-a. single-phase subway transformers. Accordingly the triple-pole units of all sizes were designed to pass through a 35 in. diameter round opening. In none of the six network cities mentioned, and together they are typical of all other cities, will it be necessary to rebuild the opening of any manhole to permit the unit to pass through; and careful investigation shows that in none of the cities where three-phase networks are being considered for the future will there be any difficulty in this regard.

If it becomes necessary to rack cables along the wall back of the network unit, it is just as easy to mount the triple-pole unit on a pipe framework braced to the wall as for the single-pole unit, since the two types are not very different in depth. However, just as with junction boxes, subway oil-circuit breakers and such apparatus, electric service companies do not consider this good practise in manholes. In only one case has a company mounted on a transformer tank a piece of auxiliary apparatus of such size as a single-pole network unit might be, and there the conditions are unlike those anywhere else in the country. In general, the place for such equipment is against a wall.

Where subsurface conditions make it impossible to build a manhole for three transformers at any particular location in the street on a main thoroughfare, three solutions have been found to be applicable: (1) obtain a vault in the basement of a building; (2) install a manhole under the sidewalk, or (3) locate the manhole in the street as close to the desired location as possible. In almost no case has it been impossible to use one of these methods. Where the third method must be employed, the distance from the most desirable location is usually so short that the cost of the extra length of duct line is small compared with the extra cost of three smaller manholes over one larger one. It is also doubtful whether city authorities will permit subsurface obstruction at three neighboring points.

The triple-pole network unit for manholes is constructed with a window of heavy, wired glass, amply strong, in the cover; this makes inspection of the principle parts easier than by taking off the cover of a single-pole unit. If it is necessary to get at the parts inside, the time to remove the few extra bolts or lugs

holding the cover on the triple-pole unit is negligible. The aluminum cover of the triple pole unit can be handled by one man without difficulty.

The design of the single-pole unit to permit easy removal of the panel from the housing when it is desired to make repairs at the shop follows the identical idea that is incorporated in the triple-pole network unit design. Similarly, right from the start, fuses have been installed on the panel of the triple-pole unit, in series with the outgoing leads, to make a separate fuse box unnecessary. These last two points, therefore, do not apply exclusively to single-pole units.

The paper claims that the single-pole unit conforms with the method of single-pole switching inherited from the radial system of distribution. In a properly designed three-phase network system single-pole switching is no longer necessary and may be abandoned. In a radial system it is better to have some light than no light, when trouble occurs on a feeder, and hence the value of single-pole switching. Networks are designed so that in no case will trouble on any feeder cause interruption of any service fed from the secondary network. Among such troubles there must always be included the putting out of service of all three conductors of a feeder, either by phase-to-phase short circuit in a three-conductor cable, or, where three single-conductor primary cables are in the same duct, by the melting down of all the conductors by a severe fault to ground on one of them. It may even be necessary to provide for the possibility of a manhole fire taking two feeders out of service simultaneously. Thus, there is no necessity in complicating the system to get the insurance that goes with keeping two of the phases in service when the third goes out. One large company even plans to use three-phase regulators on three-phase feeders serving an important network, and some are thinking of using triple-pole oil-circuit breakers at the station and three-phase distribution transformers where these can be passed through the existing manhole openings.

The paper overemphasizes the importance of manhole installations. While the difficult conditions encountered in manholes must be met and are being met, it should be remembered that of a total of over 600 triple-pole network units that will be in operation by the end of this year, less than one third will be in manholes.

Three single-pole units have three operating mechanisms in place of one for the triple-pole unit, and this means more parts to maintain. Three units together have more surface to gasket than a triple-pole unit, which gives more chance for water to leak in with any given type of construction. Three units also occupy a greater total space in the manhole, and every cubic foot is valuable. When the design of automatic network units was first under consideration, all these factors were weighed and the single-pole type was abandoned as inferior to the triple-pole type.

The simplest method of ensuring stable operation of regulators on feeders operating in parallel on a network² does not require any extra apparatus such as the transformers *T* and reactors *X* of Fig. 3. It employs only a transfer switch, corresponding to switch *S* for each feeder. This scheme was given a thorough test on the network system of the United Electric Light and Power Company, where it originated, and was shown to be entirely satisfactory.

One of the greatest problems in connection with secondary networks is the type of combined light and power secondary system to employ. Mr. Parker has blazed the trail in an effort to devise some scheme whereby the advantages of the combined system may be obtained and the disadvantages of the star connection avoided. It cannot be too strongly urged that others follow in his steps. However, it must be pointed out that the translator scheme may introduce disadvantages that outweigh those of the simple four-wire, star system, and these

2. Described in the *Electric Journal*, July, 1925.

disadvantages may so handicap the development of the network idea as to result in a loss to all concerned.

In the star system of Fig. 4 the only voltage unbalance at the motor terminals is caused by unbalance of load in the secondary mains due to varying sizes of the loads as encountered along the street. This unbalance may be reduced to a negligible amount by care in connecting two-wire and three-wire services on alternate phases. From the analysis of the translator scheme it is evident that it may easily result in voltage unbalances of at least 10 per cent at the motor terminals. For the same per cent voltage unbalance on a motor as per cent voltage reduction, both the heating and starting torque are affected to a worse degree in the case of the unbalance. Hence the effect on motors would be worse with the translator scheme than by operating them at 199 volts on a 115/199-volt star-connected secondary.

Adding extra transformers on other phases to obtain a better balance of voltage where power loads are frequent not only increases the number of transformers on the system, but also calls for larger or more manholes to house them. This would materially cut down the savings in transformer capacity gained by the diversity that networking makes possible.

Where the 115/199-volt star-connected secondary system is employed it has been necessary to use auto-transformers on not more than 15 to 20 per cent of the motors connected to the system in order to supply satisfactory voltage. On the basis of power load equal to half the lighting load these auto-transformers represent less than 7 per cent of the capacity of the secondary system. If the translators, equivalent in capacity to the total capacity of the secondary system, cause an increase of investment averaging 10 per cent of the entire distribution system cost, the auto-transformers necessary to ensure satisfactory operation of motors on a star system involve an increase in investment of less than 0.7 per cent.

The unbalance of current for the case of power load half the lighting load requires that the wire carrying the maximum current in the translator system use about 60 per cent more copper than for the star system. It would be highly inadvisable to proportion the size of the three-phase wires in the translator scheme according to the loads in those wires, because the sizes would have to be changed from block to block all over the system, and this would complicate the system even more than by adding the translators. Hence all secondary wires would be as heavy as the largest one, and this might mean a 60 per cent increase in the total amount of secondary copper required. Where 500,000-cir. mil copper is taken as the largest size, the extra copper would in many cases require a second main and duct. To these extra costs must be added those of the translators, value of man-hole space they would occupy, and losses in secondaries and translators. Even the seven-wire, separate-light-and-power secondary system would not cost more and would be simpler. The increase in cost due to all these items might easily overcome any saving gained by combining power and light mains and defeat the very purpose of the translator scheme.

If the electric service companies want to inconvenience the fewest number of customers using a star-connected system, they will adopt the 115/199-volt three-phase system, as only the polyphase motor users will be affected. In the few cases where tests show insufficient voltage, the simple auto-transformers can be employed to boost the voltage. In Memphis, a 115/199-volt system of this kind has been in operation for over ten years and the customers are entirely satisfied. In Rochester, for several years, light and power loads in large buildings downtown have been supplied by 120/208-volt transformer banks in the basements and, even though motors up to 40 h. p. are connected at the end of long risers and no auto-transformers are employed, the customers praise the service.

It must be recognized that for most systems an a-c. network system, even though fed at 13,200 volts is just a little less than

half the annual cost of a d-c. system and hardly less than 80 per cent of the cost of an a-c. radial system. These ratios have been checked independently by the engineers of five large systems. Hence, we would be deceiving ourselves as to the economic value of the translator scheme for it introduces such elements of additional cost as would wipe out the balance now in favor of a-c. networks.

The control of multiple street lamps or pole-mounting, constant-current transformers supplying series street lamps, by means of carrier current over the primary feeders, will represent a great step ahead. One company has tried out such a system, but the operation has been faulty and the relay units on the poles are of such nature as to be relatively expensive. The sender at the substation is also complicated. Another company has developed a simpler form of relay unit, and tests of a number of these, equivalent to at least a full year's service, have proved them satisfactory. This relay unit is compact, substantial and inexpensive. The sender unit is also simple and strongly built.

M. T. Crawford: Mr. Blake's paper refers to a method shown in Fig. 5 of which the title is "New Connection of Induction Regulator Circuits to Eliminate Circulating Currents." I think I am correct in stating this connection has been used recently by our company in our new Union Street substation and has proved very successful in regulating 4500-volt feeders in a multiple-feed network. One practical point in connection with it has been that some means has been found desirable to automatically disconnect the regulator control circuit on low voltage, so that regulators will not assume, during system trouble, different positions after fluctuations. I mean by that if a short circuit comes on the transmission network, and the voltage of the system as a whole oscillates back and forth, perhaps reaching very low values for brief moments, there is a tendency for the regulators to attempt to follow these voltage variations up and down. By the time matters have settled down again, some of the regulators are in one position and some in another, due to their slightly different characteristics. That has resulted at times in tripping out of network switches on the underground distribution system due to reverse flow of power for brief periods from one feeder into another where the voltage conditions were slightly different. By the simple expedient of providing a low-voltage release on the regulator control circuit, this trouble has been eliminated. The operator at the substation can reset the control circuit after the system has quieted down to normal.

The translator referred to is a very clever development. The Puget Sound Power and Light Company's underground distribution system employs single-phase, three-wire mains for lighting and alternate blocks with longitudinal alleys are placed on alternate phases, so the first alley is on one phase, the second alley on another, and so on. By that method the phases are relatively well balanced at the substation. In places, we have recently installed a fourth wire paralleling the single-phase, three-phase mains to provide small polyphase service. The result has been three-phase, four-wire mains in each alley similar to the ones referred to by Mr. Blake, and the respective alleys in the same phase relation as those shown in his diagram.

The translator, therefore, in our case offers something to look forward to as a possibility of permitting interconnection of the three sets of secondary mains for purposes of phase balancing or load protection if it should be found desirable. However, the addition of apparatus is always to be very closely scrutinized, and its necessity must be proven before it is added, as the simpler a system is, the better the service will always be.

A. H. Kehoe (communicated after adjournment): Regarding Mr. Blake's discussion of single-pole versus three-pole a-c. subway network switches I consider the proper switch to use is the one which will give minimum cost over an extended period. The cost of revising a few existing locations will be negligible on the total installation cost. Tripping and closing

elements in these switches represent a major item of cost and single-pole units will nearly triple this cost for each installation. Space, that is, cubical contents, naturally will be less in a three-pole unit than in three single units. These conditions seem to make the three-pole unit the one to be adopted generally. However, I do not favor certain of the existing three-pole switches of the so-called "battleship" construction. If the principal installation advantages set up by Mr. Blake for single-pole switches be used as specifications for three-pole switches, none of the latter's many advantages need be considered except the greatly reduced cost.

Under operating advantages there are several references to separating the phases either in the physical location or for operation of the transformers. I believe this is a case of confusing what *can* be done with what is *likely* to be done. It is possible to separate phases in three transformer vaults but it is probable that one vault three times as long will be cheaper and better if polyphase load is to be supplied. The system of polyphase secondary distribution is primarily for better universal utilization of electric service, as methods of balancing are now successful without it. However, that balanced polyphase loads are desirable at all points on a system, is a design axiom. All progress up to this time points to polyphase rather than single-phase distribution transformers for ultimate use. If the standard three-wire, single-phase system is taken as analogous to the polyphase system, it is evident that two single-phase, two-wire transformers could be placed in separate transformer vaults supplying a three-wire system. This is not likely to be found, however, in standard practise as all the capacity is found in one unit and a vault is built to take this larger unit, as this design gives minimum cost. I believe similar conditions will hold as the polyphase system develops.

Concerning the operating situation with radial distribution, the single-pole substation switches had economical advantages which could be charged to reliability. Since, in network distribution, there must be, and is always, sufficient reserve on each phase to allow for a failure of that phase, in each locality where it exists it is certain that the remaining phases will have sufficient reserve capacity to be eliminated in case of fault. Such operation does not have any effect upon service which can be equated against the increased cost, when the money, if necessary, could be expended to obtain an increased reserve upon all phases so that the particular phase in trouble would benefit by having a greater reserve. I doubt whether single-pole network switches are proper even with those systems already operating with three single-pole switches on four-wire, 4000-volt service, as new load will cause a higher voltage supply to be used than 4000 volts. It is possible to superimpose the new load on to the existing system and avoid the double transformation which is otherwise required.

Mr. Blake describes a method of cross-current compensation as the "most promising." It is, however, neither simpler nor easier of installation than others now in use. The several such connections should be given consideration before applying a definite arrangement to a particular system. The method described makes constant current and power factor the major considerations, while constant service voltage becomes a secondary one. Constant and correct service voltage is one of the most important elements in the business. While the network insures reliability and better voltage regulation than on radial systems yet this latter should not be compromised unnecessarily. It seems to me to be preferable to design the system for proper balance of load and have the regulating equipment give proper voltage under all conditions. The connection proposed does the opposite. Applying such a connection to a number of long feeders on an extended network makes other connections preferable due to the voltage variation under the ordinary load shifting.

It should be noted that a distribution network does not simulate a bus in respect to voltage or to concentration of load as indicated in Fig. 3 of the paper.

I fail to find in the translator description any reference to the use of this connection on certain distribution transformer secondaries to join and supply the different mains. If such a device ever has a practical application I believe it will be necessary to make it up in such a form, rather than as an auto-transformer, owing to cost and losses involved. Mr. Blake's statement that each section of secondary mains on a Y-connected system is not likely to be well balanced is not in accord with the purpose of adopting polyphase distribution. In some existing installations a certain amount of unbalance may exist due to the utilization having been designed for three-wire, single-phase service, but new installations are easily balanced and it is only in districts where growth of loads occur that any considerable amount of three-phase, four-wire mains are likely to be installed.

D. K. Blake: Mr. Carey questions the application of the single-pole switch. Mr. Richter's discussion also differed as to the application of the single-pole switch. It is recognized that the objection to the multiplicity of units for maintenance is valid. It is also recognized that a number of large companies will be able to utilize building and sidewalk space and, therefore, for these places the triple-pole switch is preferable. In the synopsis of the paper is this statement: "The circumstances which make single-pole switching preferable are outlined." It was my impression that there was a similar statement in the text, but on reading it over I notice there is not, and, therefore, some wrong conclusions might be drawn because of this omission. In his discussion Mr. Richter seemed to deny that these circumstances exist at all. He mentioned the cities where the large switches are being used. I agree with him that the triple-pole switch is preferable in these cities with one exception. Mr. Richter's references to the design of the switches indicates he misunderstood the paper. No exclusive features are claimed. To include all of these features in a triple-pole switch presents serious difficulties which are expensive to overcome. The operating advantages of single-pole switching may be questionable. Some engineers believe it desirable.

Mr. Richter mentions that the transfer switch shown in the *Electric Journal*, July 1925, corresponds to switch *S* shown in Fig. 3 of my paper. Switch *S* is a single-pole auxiliary switch whereas the transfer switch consists of four auxiliary switches or else one auxiliary switch to control an electrically operated double-pole double-throw switch. The cross-connected scheme is not as simple in its operation. All regulators are not adjusted at the same time according to the amount of circulating current passing through them but adjustment is obtained in a sequence. Mr. Kehoe's statement that the proposed connection does not "give proper voltage under all conditions" is not clear to the author. The line-drop compensator in each feeder maintains constant voltage at the load center with varying load. The impedances of the feeders may be different, some feeders may be long and some short. The proposed connection, by substituting a phase shifter at *T*, can also be used with three-phase regulators which is not true of other connections now in use.

I want to thank Mr. Crawford for telling us of his experience with the new regulator connection. We shall have to learn something about the operation of this connection on the systems where we use the sensitive reverse-power relay. I am glad to learn that Mr. Crawford has found a simple way of correcting the trouble.

It is not evident to the author why the translator system "may easily result in voltage unbalances of at least 10 per cent at the motor terminals." The translator system is simply the tying together of four-wire combined light-and-power mains which are supplied by delta-connected transformers. The unbalance on these mains is due to the lighting load on one phase and since this is limited to 3 per cent regulation that is also the extent of the unbalance on a 230-volt circuit. Heavy power loads are usually supplied with individual transformer banks and, there-

fore, it is not evident why larger or more manholes are necessary for this purpose on the translator system.

The translator system may be used with single-conductor cable with two of the phase wires larger than the third—just as is done in radial practise. These sizes would not have to be changed from block to block but would be uniform over the system. The case of power load equaling half the lighting load required that the wire carrying the maximum current in the translator system use about 34 per cent more copper than for the star system instead of 60 per cent as given by Mr. Richter.

Mr. Kehoe refers to a scheme using transformer secondaries. Such a connection is shown in Fig. 1, herewith, utilizing four

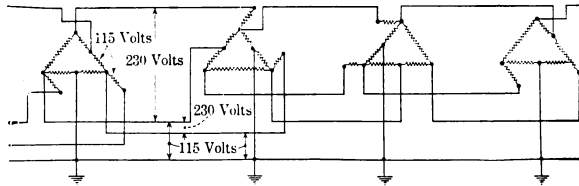


FIG. 1

standard distribution transformers. The main objection to this system is that it is tied up with the primary feeder making it necessary to provide two triple-pole network switches for protection. The translator is independent of the primary voltage. The fourth transformer is greater in kv-a. rating than the translator and very much more expensive in case of 13,200-volt or 11,000-volt primaries. The fourth transformer should be located in the same manhole with the other three while the translator may be located where convenient. Part of the lighting load is supplied from open-delta connections.

DISTRIBUTION TO SUPPLY INCREASING LOAD DENSITIES¹

(CRAWFORD)

SEATTLE, WASHINGTON, SEPTEMBER 18, 1925

C. A. Heinze: Mr. Crawford assumes as an ultimate load density in his residential areas approximately 6250 kw. per sq. mi. He has used some figures that apply apparently to Seattle, which possibly may not apply to other cities. The number of houses per block, and the number of blocks per square mile—those are local conditions. The ratio of apartment houses and flats to single residences determines the load density for residential consumers. Studies of the conditions as they exist in Los Angeles show that it is reasonable to expect future load densities of 2500 kw. per sq. mi. for single residences, 6000 kw. per sq. mi. for flats and 19,000 kw. per sq. mi. for apartment houses. The actual load density in any square mile will be determined by the ratio between the different classes.

A few years ago the Commonwealth Edison Company of Chicago had already experienced in the Loop district a load density of 75,000 kw. per sq. mi., and was preparing for a future load density in this same area of 200,000 kw. per sq. mi.

The great majority of papers presented at this convention have to do with Y-connecting the present 2300-volt systems in order to get a higher feeder voltage. Personally I think that we are wrong in stopping at this voltage. We should go higher. While some authors have intimated in their papers that if they were to start over they would probably adopt higher voltages, it seems to me the sooner we adopt the highest practical voltage considering future load densities, the easier our future distribution problems are going to be. Certainly we do not want our systems growing larger and larger with the lower voltages, and then reach a time when we have to change to a higher voltage. Why not do the changing at the time when the system is small?

One thought that seemed to run through all the papers is that

each of the authors in turn said he has a system plan. Certainly that is a big step in the right direction. The trouble in the old days was that we did not have a system plan.

Having a system plan, and realizing the enormous load densities of the future, we should build today that proportion of the ultimate plan which will give us maximum economies.

It should be realized that when a 2300-volt delta-connected system is Y-connected, we have then reached the ultimate change that can be made in that particular system. On the other hand, by adopting a higher delta voltage for the time being, we may then ultimately, when our load densities become heavier, Y-connect that and have a voltage suitable for the densities of that time.

R. E. Cunningham: I want to ask Mr. Crawford what happens when their primary phase wire falls on the ground. We have had some very serious and expensive accidents; our primary wires have fallen on the ground and lain there some time before we discovered them, and in the meantime caused troubles. To us, taking care of that feature and fully protecting the public is more important than possibly saving a little copper by using a common ground return.

Our state law requires that where the higher voltage lines such as 11-kv. and 15-kv. circuits are paralleling telephone toll lines and become grounded, we must discontinue the service until the ground is cleared. From the hazard to the public it has become a practise with us, even though it does not parallel a toll line, to pull off the circuit until the ground is cleared. On a delta-connected system we have ground detectors which will show us when our circuit is on the ground, but with a Y-connected system such as the 4-kv. we are still hunting for the answer. If Mr. Crawford could, I wish he would give us what his experience has been and what is the accepted practise of taking care of cases where the primary wire falls on the ground.

We have only recently begun the use of automatic reclosing switches having installed two smaller substations with such equipment on the 4-kv. outgoing lines. We arbitrarily adopted the practise of reclosing the first time after 2-sec. interval, the second time, 15 sec. and the third time 30 sec. If the short circuit still remains on the circuit, the switch then locks out. I would like to know what Mr. Crawford's practise is in that regard.

L. R. Gamble: I note that Mr. Crawford has planned his system somewhat in the way the telephone engineers have been planning their systems for the past several years. It always seemed to me that it was a very good scheme. However, very few companies today are doing that thing, planning for the future.

Regarding the matter of diversity in load, on distribution circuits, I note Mr. Crawford has arrived at 1600 watts per residence. In accordance with tests now being made by the National Electric Light Association, on the Electric Range Committee of which I am the engineer in charge of collecting all the data and making the reports, we have arrived at a similar figure on our more or less superficial tests. The 1600 watts per residence is on the basis of about 150 residences, and takes into account the diversity so that in considering a distribution network such as Mr. Crawford has considered here, the figure is probably fairly close. However, in radio systems, that figure is going to be somewhat higher as the diversity factor will be smaller for a smaller number of residences.

I have been more or less a dyed-in-the-wool radial fan. I have not particularly liked the network system. As time goes on, however, it seems that due to improvements in apparatus, and varied schemes in working out different problems in practise, the network system for the higher-density loads is the coming system. No doubt it is in underground systems, but in overhead systems, there seems to me to have been some sort of question.

There is one question I wanted to ask Mr. Crawford. In laying out his system for the future, he figures that he will not have

1. A. I. E. E. JOURNAL, October, 1925, p. 1063.